



4369 - A New-Generation of Synthetic Stellar Population Spectral Models Containing Stellar Binaries

Cycle: 2, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Jeff Andrews (PI)	University of Florida
Dr. Richard Henry Townsend (CoI)	University of Wisconsin - Madison
Dr. Andreas Zezas (CoI) (ESA Member)	University of Crete
Dr. Alexandros Maragkoudakis (CoI)	Bay Area Environmental Research Institute
Ms. Eirini Kasdagli (CoI)	University of Florida

OBSERVATIONS

ABSTRACT

The spectra of distant galaxies contain unique information about the star formation rate and history of the Universe at high redshift. JWST affords new opportunities to observe galaxies at high redshift as never before. Yet, our ability to uncover details about the history of the Universe are only as good as the spectral models they are compared against. The previous generation of stellar population models treat stellar binaries in only an approximate way, if at all. However, stellar mergers generate anomalously massive stars well after a starburst event, mass transfer strips the envelope off stars producing copious UV ionizing photons, and accreting compact objects producing luminous X-ray binaries. Here we propose to use the next-generation binary population synthesis code, POSYDON, to create a new generation of synthetic spectral models. POSYDON self-consistently evolves the structure of both stars, including stellar rotation, simultaneously with the orbital evolution. We propose to couple these models with detailed stellar atmospheric spectral models for both H-rich and He-rich stars, allowing us to produce a new generation of spectral models for any arbitrary star formation history and metallicity. Finally, we propose to produce a library of spectral models for integration into widely used galaxy modeling codes such as bagpipes, CIGALE, and Prospector.

OBSERVING DESCRIPTION

As this is an archival theory proposal, no observations are proposed.